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9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

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SCOUT : Supply Chain Outage Understanding & Tracker

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Introduction

Global Scenario

- Rising geopolitical tensions & climate risks
- Price spikes, shipping delays, shortages

The gap

- No real-time visibility for SMEs
- Decisions made after disruption

Our Approach

- Continuous global monitoring
- Personalized early alerts
- Proactive decision-making

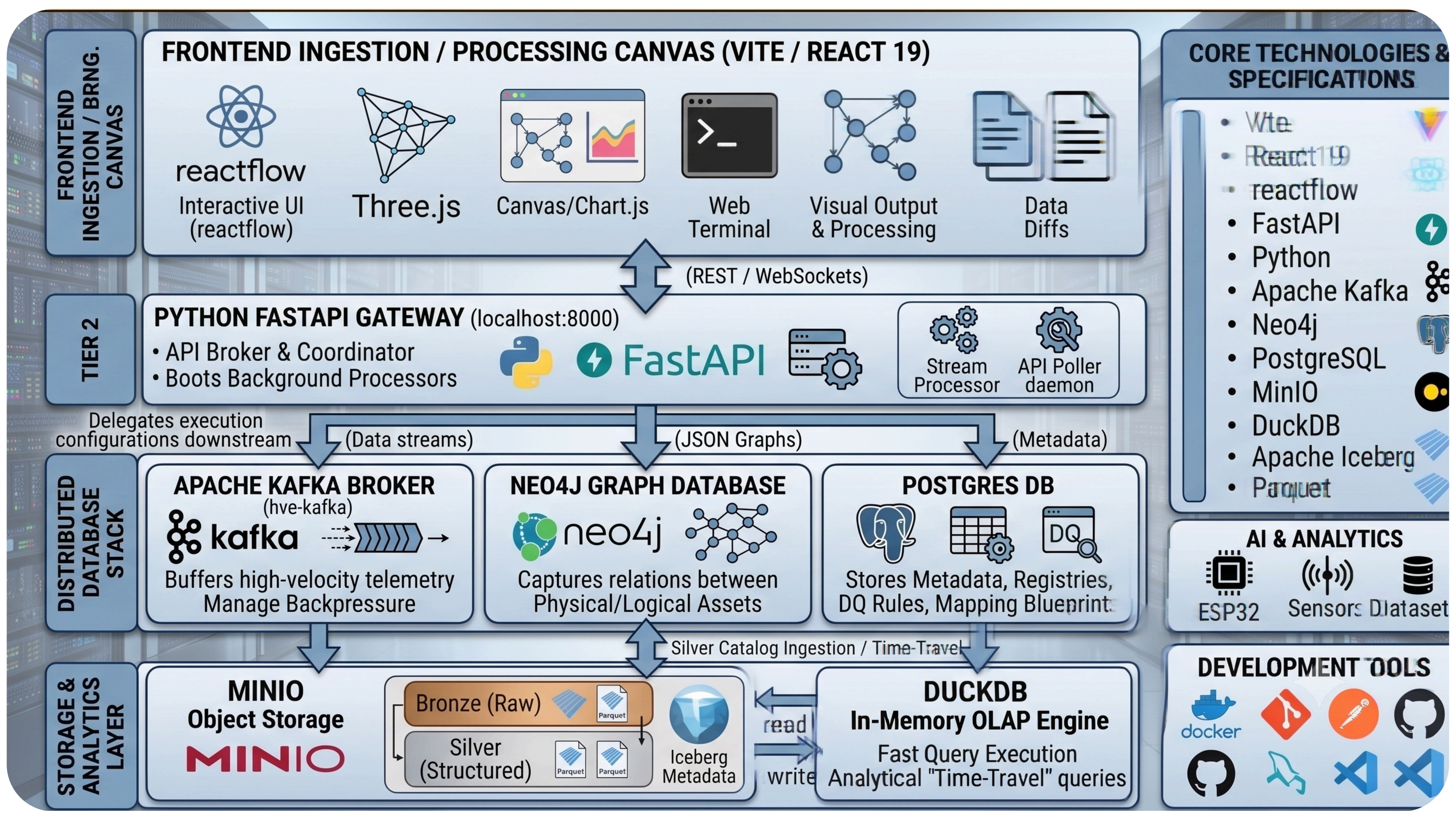
Problem Definition

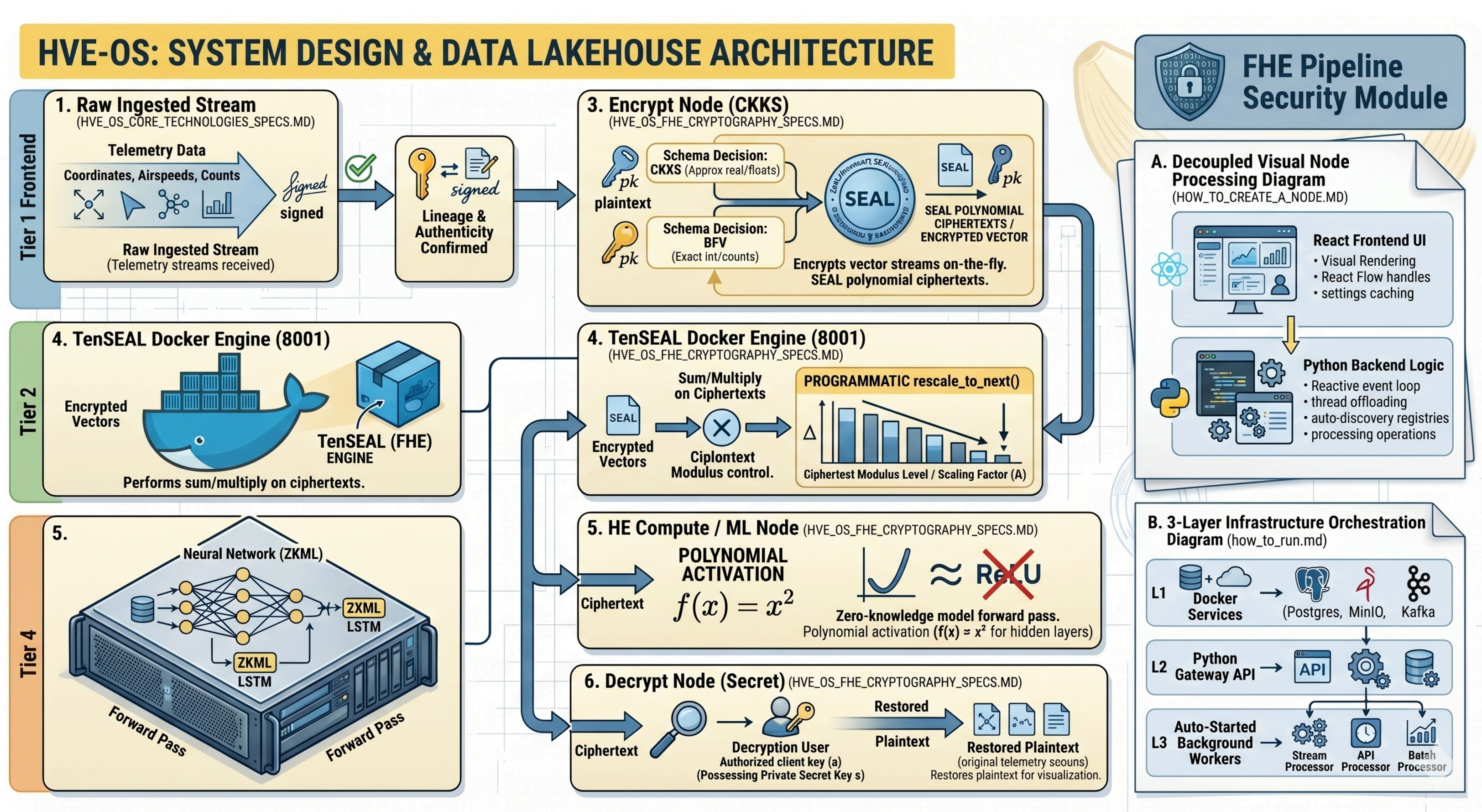
- **No Real-Time Monitoring:** Inability to track global supply chain risks continuously.
- **Fragmented Data:** Risk information is scattered across multiple sources.
- **Poor Event Detection:** Relevant disruptions are not identified automatically.
- **No Predictive Alerts:** Lack of supplier-specific risk forecasting.
- **Reactive Response:** Delayed decisions increase disruption impact.

Objectives

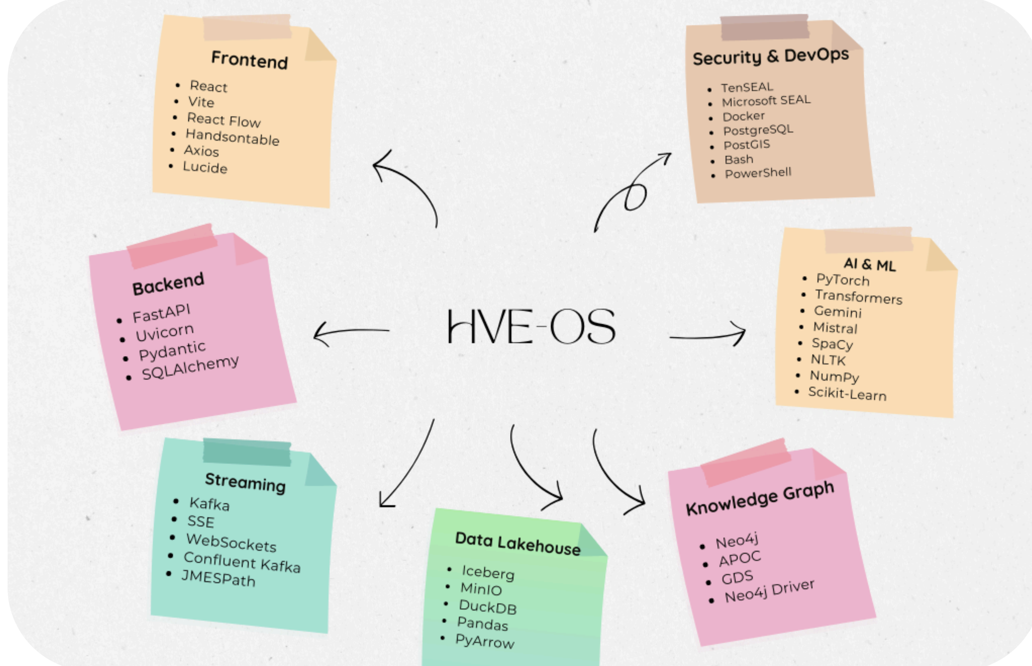
- Construct a **Multi-Source Heterogeneous** Data Ingestion Layer with Schema-Agnostic Normalisation
- Build a Property Graph **Ontology Layer** to Model Supplier Exposure Through Multi-Hop Relationship Traversal
- Develop a Weighted **Composite Risk Scoring Engine** that identifies the Risk scores,Risk Paths and overall summary
- Deliver a **Mitigation Engine** that traces all the Risk Pipelines to suggest solutions.
- Deliver a **Manufacturer-Personalised Alert Interface** including Action recommendations and Alerts.

Methodology





Tools used



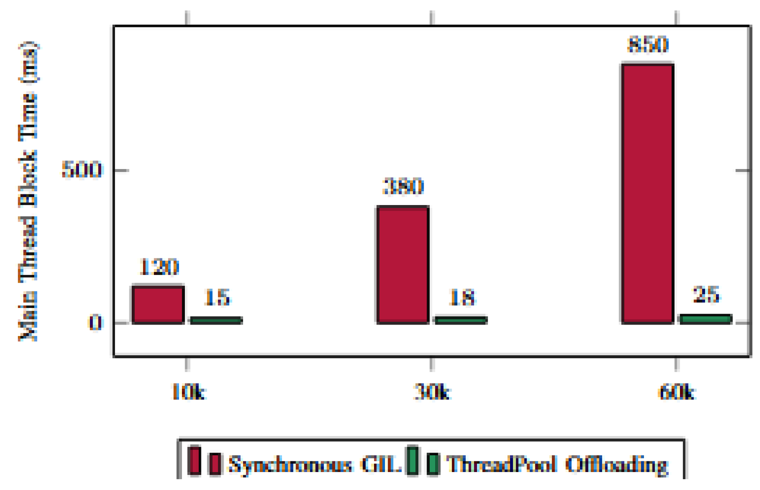
Results and Discussions

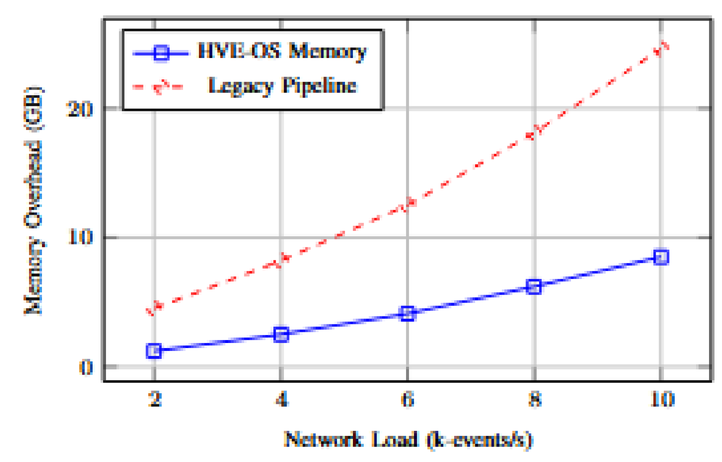
Throughput and Latency Performance:
HVE-OS achieves 5× higher throughput and 70% lower latency than traditional ETL pipelines, enabling faster real-time processing

Memory Pressure and Query Complexity:
The modular architecture significantly reduces memory overhead and scales efficiently with increasing graph complexity.

Concurrency and Thread Offloading:
Background thread-pool offloading minimizes main-thread blocking and maintains consistent system responsiveness under heavy workloads.

Risk Calculation & Routing:
The risk engine computes supplier-specific disruption scores and automatically categorizes entities for real-time risk monitoring and decision support.





Metric	Static ETL Baseline	HVE-OS (DAG)
Max Throughput (events/s)	2.1k	**10.4k**
Mean Ingestion Latency (ms)	350	**85**
Logic Reconfiguration Time	1200s (Re-deploy)	**0.5s (Hot-swap)**
AI Linking Accuracy	68%	**91%**
Graph Traversal Speed (N=4)	3.2s	**0.15s**

Conclusion

In conclusion, this proposed **Supply Chain Disruption Early Warning System** redefines how SMEs approach risk by shifting from reactive responses to anticipatory, intelligence-driven decision-making. By transforming fragmented global signals into timely, actionable insights, it empowers businesses to detect disruptions before they escalate, minimize their impact, and maintain operational continuity. Ultimately, the system aims to build resilient, adaptive supply chains capable of thriving in an increasingly uncertain and dynamic global environment.

Outcome of the work

Product Development: Developed a real-time AI-driven supply chain risk monitoring and predictive alerting system for SMEs.

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